IPhyC Qualification Round 2026

Problem A

The duration of totality is the time the shadow takes to sweep past a fixed point on the ground, so the shadow's path width is

$$w = v \cdot \Delta t.$$

The full eclipse began at 20:31:31 and ended at 20:32:32, so

$$\Delta t = 61 \text{ s}.$$

With the shadow moving at $v = 0.75 \text{ km/s}$,

$$w = 0.75 \text{ km/s} \times 61 \text{ s} = 45.75 \text{ km} \approx 46 \text{ km}.$$

Problem B

The box floats with its long face horizontal, so the 18 cm dimension is vertical and the waterline cuts across the constant cross-section $12 \times 25 \text{ cm}$.

1. Find the material volume. With wall thickness 1.0 cm (also on the bottom), the inner cavity measures

$$(18 - 2) \times (12 - 2) \times (25 - 2) = 16 \times 10 \times 23 = 3680 \text{ cm}^3,$$

while the outer volume is

$$18 \times 12 \times 25 = 5400 \text{ cm}^3.$$

So the material volume is

$$V_{\text{mat}} = 5400 - 3680 = 1720 \text{ cm}^3 = 1.72 \times 10^{-3} \text{ m}^3.$$

2. Find the mass of the box:

$$m = \rho_{\text{mat}} V_{\text{mat}} = 650 \times 1.72 \times 10^{-3} = 1.118 \text{ kg}.$$

(The air inside is negligible.)

3. By Archimedes' principle, the box displaces its own weight of water, so the displaced volume is

$$V_{\text{disp}} = \frac{m}{\rho_{\text{water}}} = \frac{1.118}{1000} = 1.118 \times 10^{-3} \text{ m}^3 = 1118 \text{ cm}^3.$$

4. With floating cross-section $A = 12 \times 25 = 300 \text{ cm}^2$, the submerged depth is

$$h_{\text{sub}} = \frac{V_{\text{disp}}}{A} = \frac{1118}{300} \approx 3.73 \text{ cm}.$$

5. The height of the top above the water surface is then

$$h_{\text{above}} = 18 - 3.73 \approx 14.3 \text{ cm}.$$

Problem C

(a) Using the given relation at $T = 20^\circ\text{C}$:

$$v = 331 + 0.60 \times 20 = 331 + 12 = 343 \text{ m/s}.$$

(b) The sound travels to the cliff and back, i.e. twice the distance d, in $t = 1.24$ s:

$$2d = v\,t = 343 \times 1.24 = 425.3 \text{ m} \quad \Rightarrow \quad d = \frac{425.3}{2} \approx 213 \text{ m}.$$

(c) At $T = 32^\circ\text{C}$ the speed of sound becomes

$$v' = 331 + 0.60 \times 32 = 331 + 19.2 = 350.2 \text{ m/s}.$$

The new echo time over the same distance is

$$t' = \frac{2d}{v'} = \frac{425.3}{350.2} \approx 1.215 \text{ s},$$

so the echo returns earlier by

$$\Delta t = 1.24 - 1.215 \approx 0.025 \text{ s} \; (\approx 25 \text{ ms}).$$

Problem E

1. During a thunderstorm, charge separation builds up between the storm cloud and the ground: the cloud base carries one sign of charge, which induces the opposite sign of charge on the ground and on structures standing on it. The building and its grounded metal rod therefore carry induced charge.

2. The electric field at the surface of a conductor is strongest where the curvature is greatest. At the sharp tip of the rod, the surface charge density is concentrated, so the local field is enormously intensified — the sharper the point, the larger the field.

3. This intense field exceeds the dielectric strength of the surrounding air and ionizes it (corona discharge). The ionized air above the rod forms a conducting channel, so a lightning leader propagates most easily from the tip. The discharge (lightning strike) therefore attaches to the rod rather than to a random, blunter part of the structure.

4. The rod is connected to the ground by a thick, low-resistance conductor. When the strike occurs, the enormous current is carried safely through this dedicated path into the ground, bypassing the building's walls, wiring, and occupants — the heating and mechanical forces of the current stay in the protection system.

5. In this way, the charge separation drives the discharge to the point where the field is concentrated, and the grounding conductor safely dissipates the strike — which is why the lightning rod protects the building.